Dynamics, the Action Principle, and the Kinematics That Supports Them

From Quaternions to the Einstein–Yang–Mills Dumbbell, Verified in Lean 4

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Preface

This book is organised around one sentence: *a shape is kinematics; putting in an action produces dynamics.* Every chapter first shows a shape, then states the action, then derives the equations, and finally points to a theorem that has been checked by the Lean 4 proof assistant.

The shape that runs through the book is the *dumbbell*: two round caps joined by a cylindrical neck. With the Einstein–Yang–Mills action the caps become closed radiation universes and the neck becomes the throat of an extremal black hole, $\mathrm{AdS}_2 \times S^2$. Both have the same size ℓ, fixed by the gauge coupling through $\ell^2 = 4\pi G/g^2$. Before we get there we build everything we need from quaternions: rotations, spin, the three-sphere, Lorentz transformations, and the geometry of relativistic momentum.

The intended reader is an undergraduate in the second to fourth year who knows linear algebra, multivariable calculus and the first courses on mechanics and electromagnetism. General relativity, quantum field theory and group theory are not assumed; we build the parts we use.

How to read the status tags

Every result carries one of the following tags, taken directly from the claim ledger of the Lean repository.

proved in Lean	checked in Lean from its stated hypotheses
external theorem	uses a theorem from the literature, cited in the text
numerical	a numerical computation, not checked by Lean
negative result	an idea that was tried and does not work
open	not solved here (and in most cases not solved anywhere)

Physical readings ("this is a gravitational wave") are kept separate from theorems and are called interpretations. Exercises stay within what Lean has checked; each exercise names the theorem that answers it.

A guide for the specialist

This section is written for readers who already know general relativity and gauge theory and want to know, in two pages, what is in this book that they have not seen before, and how far it can be trusted.

What has been checked

The book is a companion to the Lean 4 repository *Cyclic Fueter Theory* (Zenodo concept DOI 10.5281/zenodo.23448911, version 0.6.0). At that version the repository contains 1873 theorem declarations in 171 modules. An axiom audit of 2421 entries (all public theorems and the index aliases) found no `sorry`; 2396 entries depend only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`, and the rest on a subset of them. Lean verifies each theorem from its declared hypotheses. It does not verify that the hypotheses describe nature, and results quoted from the literature are tagged external theorem.

Claims of novelty

The following statements are, to the author's knowledge, not stated in this form in the literature the author has consulted. The claim is deliberately weak: no systematic review has been carried out, no specialist has confirmed it, and several items are special cases or rearrangements of known results. They are listed so that they can be checked.

(1) **Linear rigidity** (Chapter 19). Every linear perturbation of the Einstein–Yang–Mills Tolman universe that preserves spherical symmetry, the magnetic ansatz and conformal flatness is generated by a conformal Killing vector of the Einstein static universe, so it is pure gauge.

(2) **Shape of the dumbbell** (Chapter 16). Time-symmetric, conformally flat slices satisfy the Bogomolny-type inequality $r'^2 \le W^2$. The case $W = \pm r'$ saturates it and gives $r^2 + \ell^2 r'^2 = \ell^2$. The cap is the regular solution of this equation and the Bertotti–Robinson neck is its singular solution, the envelope of the caps. The junction is fixed by continuity alone, is C^1 and never C^2, and the Hamiltonian constraint admits exactly the two values 0 and $-1/\ell$ of r'' there.

(3) **Null junction** (Chapters 16–17). A timelike junction between cap and neck is impossible, and the jump $[W_u] = 1/(\sqrt{2}\,\ell)$ is transported unchanged along the null fronts.

(4) **Weyl invariant** (Chapter 16). In this sector the Weyl invariant equals $3\big(2m/r - \ell^2(1 - W^2)^2/r^2\big)$, so conformal flatness says that the Misner–Sharp mass equals the magnetic self-energy on every sphere. This is a special case of the known relation between Weyl curvature and the Misner–Sharp mass.

(5) **One pair of constraints** (Chapter 23). The same Hamiltonian and momentum constraints reduce to the Friedmann constraint on the cap and to a flat two-dimensional Klein–Gordon equation on the Kantowski–Sachs side, where the neck $r = \ell$ is a fold of minisuperspace, invariant under $r \mapsto \ell^2/r$. The neck length is canonically conjugate to $-r\dot{r}$. Minisuperspace quantisation of Reissner–Nordström interiors is known; the reading in terms of the dumbbell is what is claimed.

(6) **Constant-area data** (Chapter 26). Time-symmetric data of constant areal radius r_0 obey $W'^2 = (1 - \ell^2(1 - W^2)^2/r_0^2)/(2\ell^2)$, with the lump $W = \sqrt{2}/\cosh(x/\ell)$ at $r_0 = \ell$. Only the neck keeps its area at the next instant.

(7) **Weak gravity conjecture** (Chapter 26). With an adjoint Higgs field the electric condition for the W boson and the magnetic condition for the BPS monopole coincide, $4\pi G v^2 \leq 1$. Together with the stability bound of the neck they give $3/4 \leq (m_W \ell)^2 \leq 1$, and the neck exists only for $\Lambda \leq g^2/(16\pi G)$.

(8) **Formalisation** (whole book). The spherically symmetric Einstein–Yang–Mills–Dirac sector is machine-checked from one reduced action through the constraints, the known limits and the thermodynamic relations.

What is not claimed

The cap and neck solutions themselves, the reduced action, the canonical form and the mass function, JT gravity near the neck, black-hole thermodynamics, the sphaleron zero mode, and the weak gravity and Festina Lente conjectures are standard. The general solution of the Wheeler–DeWitt equation with matter, the quantisation of gravitational waves beyond the linear level, and any derivation of the fine-structure constant are *not* obtained; Chapter 27 lists them as open problems.

Acknowledgements

The Lean proofs, numerical checks and parts of the text were prepared with the assistance of a large language model (Claude, Anthropic); the author has checked the results and takes responsibility for the content.

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Plan of the book

This preliminary draft contains the front matter and Chapter 16, which fixes the style of the whole book. The other chapters are planned as follows; the right-hand column names the main Lean modules each chapter draws on.

	Title	Main Lean modules	Status
Part I. Quaternions and rotations			
1	From complex numbers to quaternions	`AlphaI`	planned
2	Rotations and the three-sphere	`AlphaI`, `AlphaCompleteB`	planned
3	Fueter operators	`AlphaI_Operators`, `AlphaI_C`, `AlphaI_CFamily`	planned
4	Lorentz transformations: rotation $\times$ boost	`AlphaPillar_Polar`, `AlphaRoot_Duality`	planned
Part II. The quantum sphere and the mass shell			
5	The mass-shell hyperboloid	`AlphaB_QH`, `AlphaI_H3`	planned
6	The quantum sphere	`AlphaI_QBij`	planned
7	The Dirac equation and its projector	`AlphaV`, `AlphaV_FWOrder`	planned
8	Connections and curvature: Fueter, Berry, Wigner	`AlphaI_AW`, `AlphaV_Wigner`	planned
9	Monopoles and zero modes	`AlphaCompleteC`, `AlphaCompleteD`, `ZeroModeL2`	planned
Interlude A. The Standard Model from an internal sphere			
A	$SO(10)$, Pati–Salam, $B-L$ and chirality	`AlphaVII_*`, `AlphaFix_Gauge`, `AlphaFix_Z28`	planned
Part III. Curved space and field equations			
10	Moving frames and curvature	`Frame`, `AlphaS4_Curv`	planned
11	Slices and constraints	`AlphaII`, `AlphaIII_Codazzi`	planned
12	Gauge fields	`GaugeCovariance`, `AlphaIX_YM`, `AlphaVII_Maxwell`	planned
13	The action principle	`AlphaS1_Derivation`, `AlphaS2_*`, `AlphaIX_Classical`	planned
Part IV. The dumbbell			
14	Three phases	`AlphaII_*`, `AlphaFix_Geo`, `C2`	planned
15	Cap and neck	`AlphaII_EYM`, `AlphaIII_Glue`	planned
16	The shape of the dumbbell	`AlphaDumbbell_Shape`, `AlphaDumbbell_Bogomolny`, `AlphaJunction_Unique`	**drafted**
17	Junctions	`AlphaIII*`, `AlphaS4_*`, `AlphaReal_Junction`	planned
18	Time evolution	`AlphaIV*`, `AlphaS5_*`, `AlphaReal_Rest`	planned
19	The conformal frame and Ω	`AlphaPillar_Conformal`, `AlphaOmega_*`	planned
20	Stability	`AlphaXI_*`, `AlphaPillar_Smooth`	planned
Interlude B. Contact with string theory			

	Title	Main Lean modules	Status
B	WZW, the standard embedding and attractors	`AlphaVIII_*`	planned
Part V. Heat and quanta			
21	Black-hole thermodynamics	`AlphaIX_Thermo`, `AlphaSphericalClose`	planned
22	Quantum mechanics of the neck	`AlphaIX_JT`, `AlphaFamous`	planned
23	Quantum cosmology	`AlphaCanonical`, `AlphaQuantumSpherical`, `AlphaKS_WDW`	planned
24	Quantum fields in curved spacetime	`AlphaV_FRW`, `AlphaV_Anomaly`	planned
25	Fermions	`AlphaFermion`, `AlphaV_S3Dirac`	planned
Part VI. Shape, swampland and open problems			
26	Shape degrees of freedom and the weak gravity conjecture	`AlphaGeneral`, `AlphaConstArea`, `AlphaWGC`	planned
27	Open problems	`AlphaIdeas_Check`, ledger	planned

Part I

Quaternions and rotations

Chapter 1

From complex numbers to quaternions

The question

Complex numbers rotate the plane. Is there a number system that rotates space in the same way? There is, but it needs four components instead of three, and its multiplication is not commutative. That price turns out to be exactly what rotations demand.

1.1 Why the plane is easy

Multiplying a complex number by $e^{i\theta}$ turns it through the angle θ and leaves its length alone. Most of what makes complex numbers useful in physics follows from that one fact: phases, waves, the rotation group of the plane written as a single exponential. It is natural to ask for the same thing in three dimensions, and William Rowan Hamilton spent more than a decade trying to build it out of triples $a + bi + cj$.

He could not, and there is a good reason. Rotations in space do not commute: turn a book a quarter turn about one axis and then about another, and you get a different result from doing it the other way round. A number system that rotates space must therefore have a non-commutative product. In October 1843 Hamilton saw that this can be done if one allows a fourth unit, and he scratched the rule into a bridge in Dublin:

$$i^2 = j^2 = k^2 = ijk = -1. \tag{1.1}$$

1.2 The algebra

A *quaternion* is an expression $q = a+bi+cj+dk$ with real a, b, c, d. Addition is componentwise. Multiplication is fixed by (1.1) and the distributive law, and from (1.1) alone one finds

$$ij = k, \quad jk = i, \quad ki = j, \qquad ji = -k, \quad kj = -i, \quad ik = -j.$$

The products go round the cycle $i \to j \to k \to i$ with a plus sign and the other way with a minus sign. You will meet this cycle again in Chapter 3, where it becomes the defining feature of the operators that give the theory its name.

Lean checks the four relations of (1.1) directly in Mathlib's quaternions (`i_sq`, `j_sq`, `k_sq`, `ijk`). Each proof is a line of arithmetic, which is reassuring: the whole edifice rests on four lines that nobody needs to take on trust.

It helps to split a quaternion into a real part and a vector part, $q = a + \mathbf{u}$ with $\mathbf{u} = bi + cj + dk$. For two pure quaternions the product is

$$\mathbf{u}\,\mathbf{v} = -\,\mathbf{u}\cdot\mathbf{v} + \mathbf{u}\times\mathbf{v}. \tag{1.2}$$

So the quaternion product already contains the dot product and the cross product, glued together. Historically this is where vector analysis came from; Gibbs and Heaviside later pulled the two pieces apart.

1.3 Conjugate, norm and inverse

The *conjugate* of $q = a + \mathbf{u}$ is $\bar{q} = a - \mathbf{u}$, and one checks that $q\bar{q} = \bar{q}q = a^2 + b^2 + c^2 + d^2$. We write this as $|q|^2$. The crucial property is that the norm is multiplicative.

Theorem 1.1. proved in Lean *For all quaternions p, q, $|pq|^2 = |p|^2\,|q|^2$.*

In the repository this is `normSq_mul'`.

Written out in components, the theorem says that a product of two sums of four squares is again a sum of four squares, an identity Euler had found a century before Hamilton. It also means that every nonzero quaternion has an inverse, $q^{-1} = \bar{q}/|q|^2$, so the quaternions form a division algebra: you can divide, as long as you remember on which side.

1.4 Quaternions as matrices

There is a second way to look at all of this, and it is the one physicists use most. Send the quaternion $q = a + bi + cj + dk$ to the complex matrix

$$\rho(q) = \begin{pmatrix} a + ib & c + id \\ -c + id & a - ib \end{pmatrix}. \tag{1.3}$$

Proposition 1.2. proved in Lean *The map ρ respects products, $\rho(pq) = \rho(p)\rho(q)$, it sends the conjugate to the Hermitian adjoint, $\rho(\bar{q}) = \rho(q)^\dagger$, it is injective, and*

$$\det \rho(q) = |q|^2.$$

Lean: `rho_mul`, `rho_star`, `rho_injective`, `det_rho`.

The determinant formula is the matrix version of the multiplicative norm: determinants multiply, and so do norms. Under ρ the units i, j, k become i times the Pauli matrices, up to ordering and sign. Quaternions and the spin matrices of quantum mechanics are the same object seen from two sides.

1.5 A second determinant

The matrix picture invites one more step. Take four real numbers (t, x, y, z) and form the Hermitian matrix

$$X = \begin{pmatrix} t + z & x - iy \\ x + iy & t - z \end{pmatrix}.$$

Its determinant is

$$\det X = t^2 - x^2 - y^2 - z^2,$$

which is the Minkowski interval of special relativity (`det_lorentz`). So the same 2×2 matrices carry two quadratic forms. When the matrix comes from a quaternion, the determinant is the

Euclidean length $a^2 + b^2 + c^2 + d^2$. When it is Hermitian, the determinant is the Lorentzian length $t^2 - x^2 - y^2 - z^2$.

This is the first appearance of an idea that runs through the whole book. The round sphere of Euclidean geometry and the hyperboloid of relativity are two real slices of one complex object. Chapter 4 makes this precise, and Chapter 19 shows that the dumbbell lives on both slices at once.

Where this stands

Everything in this chapter is classical and serves as the foundation. The only claim is that it has been checked: the four defining relations, the multiplicative norm, the matrix representation and both determinant formulas are theorems in the repository.

Exercises

Exercise 1.1. Use (1.1) to derive $ij = k$ and $ji = -k$. *Answer:* the relations are checked in `i_sq`, `j_sq`, `k_sq`, `ijk`.

Exercise 1.2. Write out $|pq|^2 = |p|^2|q|^2$ in components and recognise Euler's four-square identity. *Answer:* `normSq_mul'`.

Exercise 1.3. Compute $\det \rho(q)$ from (1.3). *Answer:* `det_rho`.

Exercise 1.4. Show that $\rho(q) = 0$ forces $q = 0$. *Answer:* `rho_injective`.

Exercise 1.5. Compute the determinant of the Hermitian matrix X above. *Answer:* `det_lorentz`.

Chapter 2

Rotations and the three-sphere

The question

How does a quaternion rotate a vector, and why does it take two quaternions to describe each rotation? The answer is a sandwich, $\mathbf{v} \mapsto q\mathbf{v}\bar{q}$, and the two quaternions are q and $-q$, which give the same sandwich. The set of all such q is a three-dimensional sphere, and that sphere will become the cap of the dumbbell.

2.1 The unit quaternions

A quaternion with $|q| = 1$ is a *unit quaternion*. Writing $q = a + bi + cj + dk$, the condition is $a^2+b^2+c^2+d^2 = 1$: the unit quaternions form the three-sphere S^3 sitting in four-dimensional space. Because the norm is multiplicative, the product of two unit quaternions is again a unit quaternion, and so is the inverse $\bar{q}$. The three-sphere is therefore not just a shape but a group.

Through the matrix map of Chapter 1 we can say which group.

Proposition 2.1. proved in Lean *If $|q| = 1$ then $\rho(q)$ is unitary with determinant one: $\rho(q)\rho(q)^\dagger = 1$ and $\det \rho(q) = 1$. In other words, unit quaternions are the elements of $SU(2)$.*

Lean: `rho_SU2`.

This is the identification $S^3 = SU(2)$ that every course on quantum spin uses, usually without saying so.

2.2 The sandwich

Now take a pure quaternion $\mathbf{v} = xi + yj + zk$, which we think of as an ordinary vector, and a unit quaternion q. The product $q\mathbf{v}\bar{q}$ has two properties that make it a rotation.

Proposition 2.2. proved in Lean *For all quaternions q and x,*

$$|qx\bar{q}|^2 = |q|^4\,|x|^2, \qquad \operatorname{Re}(qx\bar{q}) = |q|^2 \operatorname{Re} x.$$

Lean: `hopf_norm`, `hopf_re`.

When $|q| = 1$ the first identity says that lengths are kept, and the second says that a pure quaternion stays pure. So $\mathbf{v} \mapsto q\mathbf{v}\bar{q}$ is a length-preserving linear map of three-dimensional space. One checks that it also preserves orientation, so it is a rotation. If $q = \cos(\theta/2) + \sin(\theta/2)\,\mathbf{n}$ with a unit vector $\mathbf{n}$, the rotation is through the angle θ about $\mathbf{n}$.

Notice the half-angle. As θ runs from 0 to 2π, the quaternion only gets from 1 to -1. A full turn of space corresponds to half a trip round the sphere, and one needs a turn through 4π to

bring q back to itself. Since q and $-q$ produce the same sandwich, each rotation corresponds to exactly two unit quaternions. This is the double cover $SU(2) \to SO(3)$, and it is the reason electrons have spin one half: their states transform under q, not under the rotation it produces.

2.3 The geometry of S^3

It is worth getting a feeling for the three-sphere itself, because it is the spatial shape of the cap. Fix the real part $a = \cos\chi$ with $0 \le \chi \le \pi$. The remaining components then satisfy $b^2 + c^2 + d^2 = \sin^2\chi$, which is an ordinary two-sphere of radius $\sin\chi$. So S^3 is a stack of two-spheres that start as a point at $\chi = 0$, grow to radius one at the equator $\chi = \pi/2$, and shrink back to a point at $\chi = \pi$. Its metric is

$$d\chi^2 + \sin^2\chi\, d\Omega^2,$$

and in Chapter 16 the areal radius $r = \ell \sin(x/\ell)$ of the cap will be exactly this function with $x = \ell\chi$. The cap of the dumbbell is the northern half of this stack, cut at the equator, where the spheres are largest.

The map $q \mapsto q\, i\, \bar{q}$ goes further. It sends each unit quaternion to a unit pure quaternion, that is, to a point of the ordinary two-sphere, and the points that land on the same place form a circle. This is the Hopf fibration $S^3 \to S^2$ with circles as fibres. We will meet it again in Chapter 12 as the bundle of a magnetic monopole.

Where this stands

All of this is classical. The chapter's role is to fix two facts that are used everywhere later: unit quaternions are $SU(2)$, and S^3 is a stack of two-spheres whose radius is $\sin\chi$.

Exercises

Exercise 2.1. Show that $\rho(q)$ is unitary with determinant one when $|q| = 1$. *Answer:* `rho_SU2`.

Exercise 2.2. Show that $qx\bar{q}$ has real part $|q|^2 \operatorname{Re} x$. Why does this mean that pure quaternions are mapped to pure quaternions when $|q| = 1$? *Answer:* `hopf_re`.

Exercise 2.3. Show that $|qx\bar{q}|^2 = |q|^4|x|^2$. *Answer:* `hopf_norm`.

Exercise 2.4. Check that $\rho(1)$ is the identity matrix and $\rho(\bar{q}) = \rho(q)^\dagger$. *Answer:* `rho_one`, `rho_star`.

Chapter 3

Fueter operators

The question

Complex analysis begins with the Cauchy–Riemann operator $\partial_x + i\partial_y$, whose square root of the Laplacian makes holomorphic functions harmonic. Is there a quaternionic version, and how many are there? There are several, and the theory takes its name from one particular family of four: the *cyclic* Fueter operators. They are singled out by a simple requirement, that the four operators share a common solution.

3.1 Square roots of the Laplacian

In the plane, $(\partial_x - i\partial_y)(\partial_x + i\partial_y) = \partial_x^2 + \partial_y^2$. The Cauchy–Riemann operator is a square root of the Laplacian, and that is why the real and imaginary parts of a holomorphic function are harmonic.

Rudolf Fueter asked in the 1930s for the same thing in four dimensions with quaternion coefficients. With coordinates $x_0, \dots, x_3$ and $\partial_a = \partial/\partial x_a$, set

$$D = \partial_0 + i\partial_1 + j\partial_2 + k\partial_3, \qquad \bar{D} = \partial_0 - i\partial_1 - j\partial_2 - k\partial_3,$$

acting on quaternion-valued functions with coefficients multiplied from the left.

Theorem 3.1. proved in Lean *For commuting derivatives,*

$$\bar{D}D = D\bar{D} = \Delta_4 = \partial_0^2 + \partial_1^2 + \partial_2^2 + \partial_3^2,$$

while $D^2 = \Box + 2\partial_0(i\partial_1 + j\partial_2 + k\partial_3)$ *with* $\Box = \partial_0^2 - \partial_1^2 - \partial_2^2 - \partial_3^2$, *and* $D^2 + \bar{D}^2 = 2\Box$.

Lean: `conj_product`, `product_conj`, `square`, `square_sum`.

The last two formulas are worth a second look. The Euclidean Laplacian comes from $\bar{D}D$, but the wave operator of relativity is already hiding in D^2. If we allow a complex unit I that commutes with everything and replace the vector part by $I(i\partial_1 + j\partial_2 + k\partial_3)$, the hidden term disappears and $\bar{D}_I D_I = \Box$ exactly (`biquaternion_box`). This is the operator version of the observation at the end of Chapter 1: one algebra, two signatures.

3.2 Which coefficients work?

Any operator $\sum_a c_a \partial_a$ with quaternion coefficients can be tested in the same way. Its symbol is $\sum_a c_a \xi_a$, and the operator factorises the Laplacian when $\big(\sum_a \bar{c}_a \xi_a\big)\big(\sum_b c_b \xi_b\big) = |\xi|^2$ for every ξ.

Proposition 3.2. proved in Lean *The operator $\sum_a c_a \partial_a$ factorises the Laplacian in this sense if and only if*

$$\bar{c}_a c_b + \bar{c}_b c_a = 2\delta_{ab} \qquad \text{for all } a, b,$$

that is, if and only if (c_0, c_1, c_2, c_3) is an orthonormal basis of the quaternions.

Lean: `factorization_iff`.

So there is not one Fueter operator but many: one for every orthonormal frame. To make a definite choice one needs a further principle.

3.3 The cyclic family

Here is the principle used in this book. Instead of one operator, take a family of four, obtained from D by shifting the units round the cycle $1 \to i \to j \to k \to 1$:

$$D_m = \sum_{a=0}^{3} e_{m+a}\, \partial_a, \qquad (e_0, e_1, e_2, e_3) = (1, i, j, k),$$

with indices read modulo four. Each D_m factorises the Laplacian (`cyclic_factorizes`). The coefficients form the 4×4 quaternion matrix $C_{ma} = e_{m+a}$, and one can recover it from the operators themselves by letting them act on the coordinate functions (`coefficient_extraction`, `coefficient_unique`).

Why a cycle? Of the ways of shuffling four objects transitively with a group of order four, there are only two: the cyclic group and the Klein four-group. The two families behave very differently.

Theorem 3.3. proved in Lean *The cyclic coefficient matrix C is singular. Its right kernel is the line $\gamma(-k)\,\mathbb{H}$, where $\gamma(q) = (1, q, q^2, q^3)$, and its left kernel is $\mathbb{H}\,\gamma(i)$. Over the reals its image has dimension 12, so its quaternionic rank is 3. In contrast, the Klein family has trivial kernel.*

Lean: `right_kernel`, `left_kernel`, `rank_three`, `real_rank_twelve`, `klein_family_regular`.

The singularity has two readings, one for each kernel. The right kernel gives a common solution: for any constant quaternion q, the linear function $f = \sum_b x_b\, \gamma(-k)_b\, q$ is annihilated by all four operators at once (`null_solution`). The left kernel gives a relation among the operators themselves,

$$\sum_m \gamma(i)_m\, D_m f = 0 \qquad \text{for every } f,$$

so only three of the four are independent (`operator_relation`).

This is the precise sense in which the cyclic family is special: it is the minimal transitive family of Laplacian square roots that admits a common solution. Everything named "cyclic Fueter" in the rest of the book refers to it.

Where this stands

Fueter's operator and the factorisation of the Laplacian are classical. The derivation of the cyclic family from three requirements (factorise the Laplacian, permute the units transitively with the smallest group, share a common solution) and the computation of both kernels are part of the author's programme and are checked here. Whether the cyclic family has deeper meaning than this selection rule is an interpretation that the book does not rely on.

Exercises

Exercise 3.1. Verify $\bar{D}D = \Delta_4$ by expanding the product and using $i^2 = j^2 = k^2 = -1$ and the anticommutation of distinct units. *Answer:* `conj_product`.

Exercise 3.2. Show that $D^2 + \bar{D}^2 = 2\Box$. *Answer:* `square_sum`.

Exercise 3.3. Check that $\gamma(i) = (1, i, -1, -i)$ and $\gamma(-k) = (1, -k, -1, k)$. *Answer:* `gam_i`, `gam_mk`.

Exercise 3.4. Show that every row of C annihilates $\gamma(-k)$ from the right. *Answer:* `right_kernel_rows`.

Exercise 3.5. Show that the Klein family has no common solution. *Answer:* `klein_family_regular`.

Chapter 4

Lorentz transformations: rotation times boost

The question

Special relativity mixes space and time through boosts, while ordinary geometry only knows rotations. How are the two related? Every Lorentz transformation, written as a 2×2 complex matrix of determinant one, is a rotation followed by a boost, and in exactly one way. Rotations live on a three-sphere, boosts on a three-hyperboloid, and the two meet only at the identity.

4.1 Lorentz transformations as matrices

At the end of Chapter 1 we saw that a Hermitian 2×2 matrix X encodes a spacetime point (t, x, y, z) with $\det X = t^2 - x^2 - y^2 - z^2$. Take any complex matrix A with $\det A = 1$ and form $AXA^\dagger$.

Proposition 4.1. proved in Lean *If* $\det A = 1$ *and* X *is Hermitian, then* $AXA^\dagger$ *is Hermitian and* $\det(AXA^\dagger) = \det X$.

Lean: `lorentz_action`.

So every A in $SL(2, \mathbb{C})$ moves spacetime points while keeping the interval: it is a Lorentz transformation. As with rotations, A and $-A$ do the same thing, so $SL(2, \mathbb{C})$ is the double cover of the Lorentz group, exactly as $SU(2)$ was the double cover of the rotations.

4.2 Polar decomposition

Inside $SL(2, \mathbb{C})$ there are two kinds of special matrices. Unitary ones, $U^\dagger U = 1$, are the rotations of Chapter 2. Positive Hermitian ones, $B = B^\dagger$ with positive eigenvalues, are pure boosts: they stretch time along one direction and leave the perpendicular directions alone.

Theorem 4.2 (polar decomposition)**.** proved in Lean *Every* $M \in SL(2, \mathbb{C})$ *can be written as* $M = UB$ *with* U *unitary and* B *positive Hermitian, both of determinant one, and the two factors are unique.*

Lean: `polar_exists`, `polar_unique`.

The proof is pleasant because everything is 2×2. The matrix $M^\dagger M$ is Hermitian with determinant one (`gram_herm`, `det_gram`). For 2×2 matrices the Cayley–Hamilton theorem says

$X^2 = (\operatorname{tr} X)X - (\det X)\,1$ (`ch2`), and from it one can write down the square root explicitly:

$$B = \frac{M^\dagger M + 1}{\sqrt{\operatorname{tr}(M^\dagger M) + 2}},$$

which squares to $M^\dagger M$ (`boost_from_square`). Then $U = MB^{-1}$ is unitary. No spectral theorem is needed.

Proposition 4.3. *proved in Lean* *A matrix in $SL(2,\mathbb{C})$ that is both unitary and Hermitian with positive eigenvalues is the identity.*

Lean: `sphere_meets_hyperboloid`.

In geometric terms the rotations form a three-sphere, the boosts form a three-dimensional hyperboloid (the space of velocities), and the two touch at one point.

4.3 One matrix, two slices

There is a single formula behind all of this. For four complex numbers $z_0, \dots, z_3$ put

$$M(z) = z_0 + i\,(z_1\sigma_1 + z_2\sigma_2 + z_3\sigma_3), \qquad \det M(z) = z_0^2 + z_1^2 + z_2^2 + z_3^2.$$

When all z's are real, M is a multiple of a unitary matrix and the determinant is the Euclidean length: this is the three-sphere of rotations (`real_slice_unitary`). When z_0 is real and z_1, z_2, z_3 are imaginary, M is Hermitian and the determinant is the Minkowski length: this is the hyperboloid (`imag_slice_hermitian`). The step between them is multiplication of three components by i, the familiar Wick rotation (`wick_path`). It turns curvature $+1/R^2$ into $-1/R^2$ (`curvature_flip`).

Chapter 6 will show this in a second form. The hyperboloid of velocities can be drawn as a hemisphere, with the equator at infinite speed (`hemisphere_on_sphere`, `hemisphere_conformal`, `equator_at_infinity`). The quantum sphere of that chapter is exactly this hemisphere.

Why this matters for the dumbbell

The cap of the dumbbell is a round three-sphere (rotations) and the neck is $\mathrm{AdS}_2 \times S^2$, a space that is round in two directions and hyperbolic in the other two (rotation $\times$ boost). Chapter 19 shows that the self-dual connection $\Gamma + iK$ of general relativity splits in the same way, into a rotation part and a boost part.

Where this stands

The polar decomposition is classical. What is particular here is the explicit 2×2 proof without the spectral theorem, and the organisation of the sphere and the hyperboloid as the real and imaginary slices of one determinant, which the book uses repeatedly.

Exercises

Exercise 4.1. Prove the Cayley–Hamilton identity $X^2 = (\operatorname{tr} X)X - (\det X)1$ for 2×2 matrices. *Answer:* `ch2`.

Exercise 4.2. Show that $\det(M^\dagger M) = 1$ when $\det M = 1$. *Answer:* `det_gram`.

Exercise 4.3. Check that $B = (M^\dagger M + 1)/\sqrt{\operatorname{tr}(M^\dagger M) + 2}$ squares to $M^\dagger M$. *Answer:* `boost_from_square`.

Exercise 4.4. Compute $\det M(z)$ for the matrix above. *Answer:* `det_M`.

Exercise 4.5. Show that $1/\cosh\eta \le 2/(\eta+1)$ for $\eta \ge 0$, and explain why it puts the equator of the hemisphere at infinite rapidity. *Answer:* `equator_at_infinity`.

Part II

The quantum sphere and the mass shell

Chapter 5

The mass-shell hyperboloid

The question

What shape does relativistic momentum have? For a particle of mass m the energy and momentum satisfy $E^2 - p^2 = m^2$, which is a hyperboloid. This chapter learns to measure on it, with rapidity as the natural coordinate, and meets the Gudermann function that will reappear in the quantum sphere and on the neck of the dumbbell.

5.1 Velocity space

A particle of mass $m > 0$ and momentum $\mathbf{p}$ has energy $E = \sqrt{m^2 + |\mathbf{p}|^2}$. Divide by m and write $X_0 = E/m$ and $\mathbf{u} = \mathbf{p}/m$; then

$$X_0^2 - |\mathbf{u}|^2 = 1, \qquad X_0 > 0.$$

This is one sheet of a hyperboloid in four dimensions. Its points are four-velocities, so this three-dimensional surface is the space of all possible states of motion of the particle. It is the boost half of Chapter 4: every point is reached from the rest point $(1, \mathbf{0})$ by exactly one boost.

The natural coordinate along a radial line is the *rapidity* η, defined by $|\mathbf{p}| = m \sinh \eta$ and $E = m \cosh \eta$. Velocities do not add simply, but rapidities along one line do, which is why η rather than v measures distance on the hyperboloid.

5.2 The geometry of the hyperboloid

With the metric induced from Minkowski space, the hyperboloid of radius m has

$$ds_H^2 = m^2 \big(d\eta^2 + \sinh^2 \eta \, d\Omega^2 \big),$$

which is hyperbolic space of curvature $-1/m^2$. A short computation along a radial line shows what this means in practice.

Proposition 5.1. proved in Lean *On the mass shell $E(p) = \sqrt{m^2 + p^2}$ the slope is $dE/dp = p/E = v$, the velocity. The outward normal in the Minkowski sense is $(E/m, p/m)$, the four-velocity, and it has Minkowski length -1. At $p = 0$ the second derivative of $X_0(u) = \sqrt{1 + u^2}$ is $+1$.*

In the repository: `slope_H`, `normal_H`, and the curvature at the contact point inside `cayley_klein`.

So the shell bends upward away from the rest point, and the normal to the shell at any point is the four-velocity of the particle in that state. Integrating, a disc of rapidity radius η_0

in a two-dimensional slice of the shell has area

$$2\pi m^2(\cosh\eta_0 - 1) = 2\pi m\,(E_0 - m),$$

proportional to the kinetic energy at its rim (`area_H`). Its curvature and the geodesic curvature of its boundary obey the Gauss–Bonnet theorem, as they must (`gauss_bonnet_H`).

5.3 The Gudermann function

One function connects rapidity to angles, and it will turn up again and again. It is the Gudermann function

$$\operatorname{gd}\eta = \arctan(\sinh\eta), \qquad \frac{d}{d\eta}\operatorname{gd}\eta = \frac{1}{\cosh\eta}.$$

Proposition 5.2. proved in Lean *If $\chi = \operatorname{gd}\eta$, then*

$$\sin\chi = \tanh\eta = v, \qquad \cos\chi = \frac{1}{\cosh\eta} = \frac{1}{\gamma}, \qquad \tan\chi = \sinh\eta = \frac{p}{m}.$$

The function gd *is strictly increasing and maps the real line onto $(-\pi/2, \pi/2)$.*

In the repository: `gudermann`, `gudermann_momentum`, `gd_deriv`, `gd_strictMono`, `gd_surj`.

In words: an angle χ that runs only up to a right angle can record every rapidity from zero to infinity. The cosine of that angle is the inverse Lorentz factor and the sine is the velocity. Infinite speed sits at $\chi = \pi/2$, at a finite angle. Chapter 6 turns this into a picture, and in Chapter 17 the same function will describe the arrival of a wave front on the neck of the dumbbell.

Where this stands

Classical. The chapter's job is to fix rapidity and the Gudermann function as the coordinates in which the quantum sphere and the mass shell will be compared.

Exercises

Exercise 5.1. Show that $dE/dp = p/E$. *Answer:* `slope_H`.

Exercise 5.2. Show that the Minkowski normal of the shell is the four-velocity and has length -1. *Answer:* `normal_H`.

Exercise 5.3. Prove $\sin(\operatorname{gd}\eta) = \tanh\eta$ and $\cos(\operatorname{gd}\eta) = 1/\cosh\eta$. *Answer:* `gudermann`.

Exercise 5.4. Differentiate gd. *Answer:* `gd_deriv`.

Exercise 5.5. Compute the area of a disc of rapidity radius η_0 on the shell. *Answer:* `area_H`.

Chapter 6

The quantum sphere

The question

Quantum mechanics describes a free relativistic particle by a state that depends on its momentum. Where do these states live? Normalised, they fill a hemisphere of a three-sphere, the *quantum sphere*. It touches the mass shell of Chapter 5 at the rest point and curves the opposite way, and its height is the reciprocal of the shell's height.

6.1 The map Q

For $m > 0$ and $\mathbf{p} \in \mathbb{R}^3$ put

$$Q(\mathbf{p}) = \Big(\frac{m}{E}, \frac{\mathbf{p}}{E}\Big), \qquad E = \sqrt{m^2 + |\mathbf{p}|^2}.$$

The four numbers satisfy $(m/E)^2 + |\mathbf{p}/E|^2 = 1$, so $Q(\mathbf{p})$ is a unit quaternion, a point of S^3, and its first component is positive (`qD_unit`, `qD_upper`). In matrix language it is the unitary matrix $U(\mathbf{p}) = (m - i\boldsymbol{\sigma}\cdot\mathbf{p})/E$, which appears in the free Dirac equation; Chapter 7 explains why.

Theorem 6.1. proved in Lean *Q is a bijection from $\mathbb{R}^3$ onto the open upper hemisphere $\{(w, \mathbf{x}) : w > 0,\ w^2 + |\mathbf{x}|^2 = 1\}$, with inverse $\mathbf{p} = m\mathbf{x}/w$.*

In the repository: `qmap_bij`, with `qmap_inj`, `qmap_surj`, `qinv_left`, `qinv_right`.

Every momentum gives a point of the hemisphere, and every point is reached once. Rest is the north pole. Momenta of ever larger size crowd towards the equator, which they never reach: the equator is infinite momentum.

6.2 Reciprocal heights

Put the quantum sphere and the mass shell side by side, in the coordinates of Chapter 5. The shell point is $(X_0, \mathbf{u}) = (E/m, \mathbf{p}/m)$ and the sphere point is $(w, \mathbf{x}) = (m/E, \mathbf{p}/E)$.

Proposition 6.2. proved in Lean *$wX_0 = 1$ and $\mathbf{x} = \mathbf{u}/X_0$. The polar angle of Q is the Gudermann angle of the rapidity, $\chi = \operatorname{gd}\eta$.*

In the repository: `qh_reciprocal`, `gudermann`.

So the two surfaces are mirror images through the plane of Newtonian physics. Both pass through the rest point with zero slope, the shell bends up with second derivative $+1$ and the sphere bends down with -1. They belong to a single family $X_0 = \sqrt{1 - \kappa X^2}$, all tangent at the rest point, with second derivative $-\kappa$: the sphere is $\kappa = 1$, the shell $\kappa = -1$, and Newton is $\kappa = 0$ (`cayley_klein`). Restoring units, $\kappa = \pm 1/c^2$.

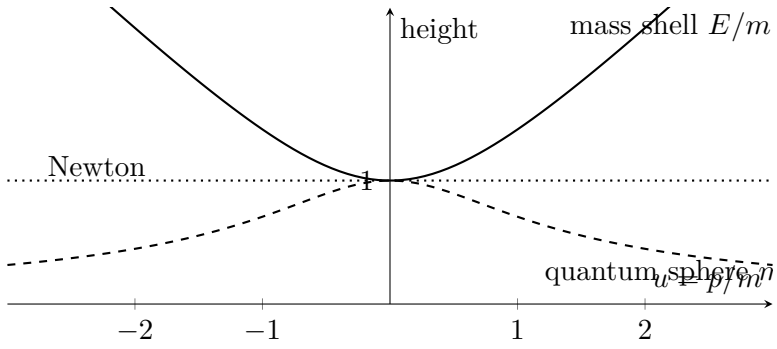


Figure 6.1: The mass shell (solid) and the quantum sphere (dashed) over the same proper velocity u. They touch at rest with the Newtonian plane (dotted) and bend in opposite directions; their heights are reciprocal.

6.3 The metric of the quantum sphere

The quantum sphere is not just a picture; it has a metric, induced from S^3, and it is related to the hyperbolic metric of the shell by a simple factor.

Proposition 6.3. proved in Lean *The metric of S^3 pulled back by Q is*

$$\partial_i Q \cdot \partial_j Q = \frac{\delta_{ij}}{E^2} - \frac{p_i p_j}{E^4},$$

and it equals the metric of the mass shell divided by E^2: the quantum hemisphere is conformal to hyperbolic space.

In the repository: `pullback_metric`, `conformal`, `metric_pullback`.

The conformal factor has a direct consequence for areas. A cap of the hemisphere of polar radius $\chi_0 = \operatorname{gd} \eta_0$ has area $2\pi(1 - m/E_0)$, and the corresponding disc on the shell has area mE_0 times larger (`area_Q`, `area_ratio`). As $E_0 \to \infty$ the hemisphere's area tends to the finite value 2π while the shell's grows without bound. Infinite momentum space has been folded into a finite, round region, and it is round in the way that rotations are round.

A thought to carry forward

In Chapter 4 the sphere of rotations and the hyperboloid of boosts were the real and imaginary slices of one complex quadric. Here the same pair appears in physics: the hyperboloid is where the particle's motion lives, and the hemisphere is where its quantum state lives. The cap of the dumbbell will be a round S^3, and the neck will contain a hyperbolic plane.

Where this stands

The map Q and its pullback metric are elementary, and the relation to the Dirac Hamiltonian is standard. The organisation of shell, sphere and Newtonian plane into one family tangent at rest, and the bookkeeping of areas, are presentations developed for this book; they are checked but not claimed as research results.

Exercises

Exercise 6.1. Show that $Q(\mathbf{p})$ is a unit quaternion with positive real part. *Answer:* `qD_unit`, `qD_upper`.

Exercise 6.2. Show that Q is injective. *Answer:* `qmap_inj`.

Exercise 6.3. Show that $wX_0 = 1$. *Answer:* `qh_reciprocal`.

Exercise 6.4. Compute the second derivatives of $\sqrt{1+u^2}$ and $1/\sqrt{1+u^2}$ at $u = 0$. *Answer:* `curv_H_Q`, `cayley_klein`.

Exercise 6.5. Compute the area of a cap of the hemisphere. *Answer:* `area_Q`.

Chapter 7

The Dirac equation and its projector

The question

Why does the unit quaternion $U(\mathbf{p}) = (m - i\boldsymbol{\sigma} \cdot \mathbf{p})/E$ of Chapter 6 deserve the name "quantum"? Because it is what the Dirac equation produces when one asks for the states of positive energy. The projector onto those states has a natural metric, and that metric is exactly half the metric of the quantum sphere.

7.1 A square root of E^2

Dirac looked for a Hamiltonian linear in momentum whose square is $E^2 = m^2 + |\mathbf{p}|^2$. Writing $H = m\beta + \alpha_1 p_1 + \alpha_2 p_2 + \alpha_3 p_3$ and squaring, the cross terms must cancel and the squares must be one.

Proposition 7.1. proved in Lean *If $\beta^2 = \alpha_i^2 = 1$ and $\beta, \alpha_1, \alpha_2, \alpha_3$ anticommute pairwise, then*

$$H^2 = (m^2 + p_1^2 + p_2^2 + p_3^2)\, 1.$$

In the repository: `dirac_sq`. The proof holds in any associative algebra, so it does not care which matrices one uses.

Compare this with Chapter 3. There we wanted a first-order operator whose "square" is the Laplacian, and quaternion units did the job. Here we want one whose square is E^2, and anticommuting matrices do the job. The two problems are the same problem, and in the standard representation the α_i are built from the quaternion units of Chapter 1.

7.2 The positive-energy projector

Since $H^2 = E^2$, the eigenvalues of H are $\pm E$, and

$$P_+ = \tfrac{1}{2}\Big(1 + \frac{H}{E}\Big)$$

is the projector onto positive energy.

Proposition 7.2. proved in Lean *If $H^2 = E^2$ and $E \neq 0$, then $P_+^2 = P_+$.*

In the repository: `proj_idem`.

As $\mathbf{p}$ varies, $P_+(\mathbf{p})$ traces out a family of two-dimensional subspaces, and any such family carries a metric, the *quantum metric* $\frac{1}{2}\operatorname{tr}(\partial_i P_+\, \partial_j P_+)$. It measures how fast the positive-energy states change with momentum.

Theorem 7.3. proved in Lean *The quantum metric of P_+ equals one half of the metric pulled back by the map Q of Chapter 6:*

$$\tfrac{1}{2}\operatorname{tr}(\partial_i P_+\,\partial_j P_+) = \tfrac{1}{2}\Big(\frac{\delta_{ij}}{E^2} - \frac{p_i p_j}{E^4}\Big).$$

In the repository: `quantum_metric`, together with `pullback_metric`.

So the space of positive-energy Dirac states, measured in its own natural way, is a hemisphere of radius $1/\sqrt{2}$. Along a radial line the polar angle is $\chi = \arctan(p/m)$, whose derivative is m/E^2 (`chi_deriv`, `radial_metric`). The "quantum sphere" is therefore not a metaphor: it is the geometry of Dirac states.

7.3 The non-relativistic limit

The Dirac equation should reduce to the Schrödinger–Pauli equation at low speed. The standard way to see this is the Foldy–Wouthuysen transformation, which removes the terms that mix upper and lower components order by order in $1/m$. Write $H = m\beta + \mathcal{O} + \mathcal{E}$ with an odd part $\mathcal{O}$ (anticommuting with β) and an even part $\mathcal{E}$.

Theorem 7.4. proved in Lean *After one transformation with generator $\beta\mathcal{O}/(2m)$ the Hamiltonian is, through order $1/m^2$,*

$$\beta\Big(m + \frac{\mathcal{O}^2}{2m}\Big) + \mathcal{E} - \frac{1}{8m^2}[\mathcal{O},[\mathcal{O},\mathcal{E}]],$$

plus odd terms of order $1/m$. A second transformation removes those odd terms without changing the even part at this order.

In the repository: `fw_order2`, `fw_hadamard3`, `second_fw`, `odd1_factor`.

With $\mathcal{O} = \boldsymbol{\alpha}\cdot\boldsymbol{\pi}$ in an external electromagnetic field, $\mathcal{O}^2/2m$ becomes the Pauli Hamiltonian with its $g = 2$ magnetic moment, and the double commutator produces the Darwin term and the spin–orbit coupling (`pauli_nc`, `darwin_form`, `so_form`). The algebra is exact; the expansion itself is a formal series in $1/m$, which the source states explicitly.

Where this stands

The Dirac equation, the projector and Foldy–Wouthuysen are textbook material. The identification of the quantum metric with half the pullback metric of Q is the bridge that the rest of Part II stands on; it is a short computation, checked here, and the author has not found it stated in this form.

Exercises

Exercise 7.1. Square $m\beta+\boldsymbol{\alpha}\cdot\mathbf{p}$ using only the anticommutation relations. *Answer:* `dirac_sq`.

Exercise 7.2. Show that P_+ is idempotent. *Answer:* `proj_idem`.

Exercise 7.3. Show that $\big(-mp/E^3\big)^2 + \big(m^2/E^3\big)^2 = (m/E^2)^2$. What does it say about the radial direction? *Answer:* `radial_metric`.

Exercise 7.4. Differentiate $\arctan(p/m)$. *Answer:* `chi_deriv`.

Chapter 8

Connections and curvature: Fueter, Berry, Wigner

The question

Carry a spinning particle round a loop in momentum space and bring it back. Does its spin come back the same? Not necessarily, and there are three natural ways to say by how much: the Fueter frame, the Berry phase of quantum mechanics, and the Wigner rotation of relativity. Two of them turn out to be the same connection in different gauges. The third is genuinely different, and one number tells them apart.

8.1 What a connection is

A connection tells you how to compare things at nearby points. In our setting the "things" are two-component spinors attached to each momentum $\mathbf{p}$, and a connection is a matrix-valued one-form $A = \sum_i A_i\, dp_i$ with A_i in $\mathfrak{su}(2)$. Moving a spinor along a path with $\dot\psi = -A(\dot{\mathbf{p}})\psi$ is *parallel transport*. Its failure to return round a small loop is measured by the curvature $F = dA + A \wedge A$.

Two connections are *gauge equivalent* if they differ only by a change of basis at each point, $A' = gAg^{-1} + g\, d(g^{-1})$. Then $F' = gFg^{-1}$, so anything built from F that is unchanged by conjugation is a gauge invariant.

8.2 Three connections

Using $U(\mathbf{p}) = (m - i\boldsymbol{\sigma}\cdot\mathbf{p})/E$ from Chapter 6:

- the *Fueter* (frame) connection is $A_F = \frac{1}{2}U^{-1}dU$;
- the *Berry* connection of positive-energy Dirac spinors is $A_B = \dfrac{i\boldsymbol{\sigma}\cdot(\mathbf{p}\times d\mathbf{p})}{2E(E+m)}$;
- the *Wigner* connection, the rotation part of the standard boost $L(\mathbf{p}) = \big((E+m) + \boldsymbol{\sigma}\cdot\mathbf{p}\big)/\sqrt{2m(E+m)}$, is $A_W = -\dfrac{i\boldsymbol{\sigma}\cdot(\mathbf{p}\times d\mathbf{p})}{2m(E+m)}$.

One reads off at once that $A_B = -(m/E)A_W$. They point the same way but have different strengths, and the ratio depends on $\mathbf{p}$. That alone does not prove they are inequivalent, because gauge transformations can do surprising things. One needs an invariant.

The Wigner rotation deserves its own sentence. For a Lorentz transformation Λ, the matrix $W = L(\Lambda\mathbf{p})^{-1}\Lambda L(\mathbf{p})$ is a rotation: unitary with determinant one (`wigner_unitary`, `wigner_det`), and it composes correctly, $W(\Lambda_2\Lambda_1, \mathbf{p}) = W(\Lambda_2, \Lambda_1\mathbf{p})W(\Lambda_1, \mathbf{p})$ (`wigner_cocycle`). Along a path of pure boosts it is transported exactly by A_W (`transport`), so Thomas precession is the holonomy of this connection.

8.3 The invariant that separates them

For any connection of the form $A_i = a(s)\, i\boldsymbol{\sigma}\cdot(\mathbf{p}\times e_i) + b(s)\, i\sigma_i$ with $s = |\mathbf{p}|^2$, one can compute the curvature in closed form and then the trace-Gram invariant $\mathcal{G} = \sum_k -2\,\mathrm{tr}(F_k^* F_k^*)$, which is unchanged by gauge transformations (`curv_formula`, `gram_trace`, `gram_gauge`).

Theorem 8.1. proved in Lean *For the three connections above,*

$$\mathcal{G}_F = \mathcal{G}_B = \Big(\frac{m}{E}\Big)^4 \mathcal{G}_W, \qquad \mathcal{G}_W = \frac{2}{m^2E^2} + \frac{1}{m^4}.$$

Hence for $E > m > 0$, that is for every nonzero momentum, $\mathcal{G}_B < \mathcal{G}_W$, and the Berry and Wigner connections are not gauge equivalent. The Fueter and Berry connections are gauge equivalent, with the gauge transformation given by the Foldy–Wouthuysen half rotation $V = \big((E+m) - i\boldsymbol{\sigma}\cdot\mathbf{p}\big)/\sqrt{2E(E+m)}$, which satisfies $V^2 = U$.

In the repository: `trG_W`, `trG_B`, `trG_F`, `ratio`, `berry_not_gauge_wigner`, `half_rotation_sq`, `berry_fueter_gauge`.

The pleasant surprise is the last sentence. The connection defined purely from the quaternion U (Fueter) and the one defined from quantum states (Berry) are the same object, and the change of gauge between them is the transformation that separates particles from antiparticles at low energy. The relativistic Wigner rotation is the odd one out, by a factor $(m/E)^4$ in the invariant. In the half-angle form, $A_B = \frac{1}{2}(1 - m/E)\, i\boldsymbol{\sigma}\cdot(\mathbf{n}\times d\mathbf{n})$ and $A_W = -\frac{1}{2}(E/m - 1)\, i\boldsymbol{\sigma}\cdot(\mathbf{n}\times d\mathbf{n})$ with $\mathbf{p} = |\mathbf{p}|\mathbf{n}$ (`berry_half_angle`, `wigner_half_angle`).

Where this stands

Berry phases of Dirac spinors and Thomas–Wigner rotation are each well known, and the relation $A_B = -(m/E)A_W$ is easy. The gauge-invariant proof that they are inequivalent, and the identification of Berry with Fueter through the Foldy–Wouthuysen half rotation, are results of the author's programme; they are checked here and the author has not found them in this form. A separate paper on the observable consequences of the difference between A_B and A_W is in preparation.

Exercises

Exercise 8.1. Show that the Wigner rotation is unitary with determinant one. *Answer:* `wigner_unitary`, `wigner_det`.

Exercise 8.2. Verify the cocycle rule. *Answer:* `wigner_cocycle`.

Exercise 8.3. Show that $\mathcal{G}$ is unchanged when F is replaced by gFg^{-1}. *Answer:* `gram_gauge`.

Exercise 8.4. Show that $V^2 = U$. *Answer:* `half_rotation_sq`.

Exercise 8.5. Show that $\mathcal{G}_B < \mathcal{G}_W$ whenever $E > m > 0$. *Answer:* `trG_B_lt`.

Chapter 9

Monopoles and zero modes

The question

A magnetic monopole is the simplest place where geometry forces a quantum state to exist. Put a charged spinning particle on a sphere around a monopole of charge q, and there are exactly $2q$ states of zero energy, all of one handedness. In this chapter the monopole appears twice: once in momentum space, where the Berry connection of Chapter 8 is a monopole, and once on an ordinary sphere, which will later be the cross-section of the neck.

9.1 A monopole in momentum space

The Berry connection of Chapter 8 lives on momentum space, but Chapter 6 taught us that momentum space carries a natural curved metric. Take the metric $h = (\delta_{ij} - p_i p_j/E^2)/m^2$. Along a radial line its length element is dp/E, and across it the length element is p/m times the round one (`h_radial`, `h_tangential`). Writing $p = m\sinh\eta$ turns this into

$$h = d\eta^2 + \sinh^2\eta\, d\Omega^2,$$

hyperbolic space with curvature exactly -1 (`h_rapidity`, `h3_curvature`).

Theorem 9.1. *proved in Lean* *On this hyperbolic space the Berry connection A_B and the Higgs field $\Phi = (\mathbf{p}/E)\cdot T$, with $T = -i\boldsymbol{\sigma}/2$, satisfy the Bogomolny equation*

$$D_A\Phi = -{*}F.$$

In the repository: `starF_closed` and its supporting lemmas in `AlphaCompleteC`.

So the Berry phase of a relativistic electron is not just any connection: it is a BPS monopole on the space of velocities. Its Higgs field is the velocity $\mathbf{p}/E$ itself, which goes from zero at rest to unit length at infinite speed, the equator of the quantum sphere. We will meet the Bogomolny equation again in Chapter 16, where an inequality of exactly this type governs the shape of the dumbbell.

9.2 Zero modes on a sphere

Now put a particle of charge q on a unit two-sphere around a monopole. After separating variables, the zero-energy condition for the Dirac operator reduces to a first-order equation in the polar angle,

$$u' = V(\theta)\,u, \qquad V = \frac{m + q\cos\theta}{\sin\theta},$$

with m the azimuthal quantum number. It can be solved exactly.

Proposition 9.2. proved in Lean *The solution is* $u(\theta) = \tan^m(\theta/2)\,\sin^q\theta$. *Near* $\theta = 0$ *it behaves like* θ^{m+q} *and near* $\theta = \pi$ *like* $(\pi-\theta)^{q-m}$, *so it is square integrable exactly when* $|m| < q + \frac{1}{2}$.

In the repository: `zero_mode`, `exponent_condition`, `zero_mode_L2`.

Counting the allowed values of m gives the index.

Theorem 9.3. proved in Lean *If* q *is an integer,* m *runs over* $2q$ *half-integers; if* q *is a half-integer, over* $2q$ *integers. In both cases there are exactly* $2q$ *zero modes, and the other chirality has none.*

In the repository: `count_integer_charge`, `count_half_charge`, `other_chirality_empty`.

This is the Atiyah–Singer index theorem in its simplest instance, derived here by hand: the number of zero modes equals the magnetic flux, and they all have the same handedness. One also needs to know that the other solutions of the radial equation are really not normalisable, not just that their leading terms are not; that is a short estimate (`large_not_L2`).

Looking ahead

The neck of the dumbbell is $\mathrm{AdS}_2 \times S^2$, and the Yang–Mills field on it is exactly a monopole of charge $1/2$ on the S^2 for each isospin component. By Theorem 9.3 each component carries one zero mode, which becomes a massless fermion in the two-dimensional AdS_2 (Chapter 25).

Where this stands

Monopole zero modes and their counting are classical. That the Berry connection of Dirac spinors, together with the velocity as a Higgs field, solves the Bogomolny equation on the space of velocities is part of the author's programme; it is checked here component by component, including the curvature -1 of the stage.

Exercises

Exercise 9.1. Show that the metric h becomes $d\eta^2 + \sinh^2\eta\, d\Omega^2$ under $p = m\sinh\eta$. *Answer:* `h_rapidity`.

Exercise 9.2. Check that $u = \tan^m(\theta/2)\sin^q\theta$ solves $u' = Vu$. *Answer:* `zero_mode`.

Exercise 9.3. Count the zero modes for $q = 1$ and for $q = 3/2$. *Answer:* `count_integer_charge`, `count_half_charge`.

Exercise 9.4. Show that a power θ^e is square integrable near 0 exactly when $e > -1/2$. *Answer:* `exponent_condition`.

Interlude A

The Standard Model from an internal sphere

The question

Can the quaternionic picture say anything about the particles we actually see? It can say something about their charges. A round nine-sphere has the symmetry $SO(10)$; stretching it into an ellipsoid breaks the symmetry to the Pati–Salam group, and the lowest Dirac modes carry exactly the charges of one generation of quarks and leptons. The dynamics that would select this ellipsoid is not derived, and this interlude says so clearly.

This interlude can be skipped without loss. It is placed here because it uses only the Dirac operator and the monopole of the preceding chapters, and because it is the part of the programme that touches particle physics most directly.

A.1 Why one $SU(2)$ is not enough

The quantum sphere of Chapter 6 is a copy of $SU(2)$. Electroweak physics needs $SU(2)_L \times U(1)_Y$, where the hypercharge Y commutes with all of $SU(2)_L$. Inside a single $\mathfrak{su}(2)$ that is impossible: a pure quaternion commuting with both i and j is zero (`center_pure_trivial`). With no Y the weak mixing angle would satisfy $\sin^2\theta_W = 1$ (`no_mixing`). This is a genuine obstruction, and the programme records it as a negative result. negative result

The way out is to use both sides. Left and right multiplication of quaternions commute with each other (`left_right_commute`), giving $SU(2)_L \times SU(2)_R$, and the inversion $U \mapsto U^{-1}$, which is momentum reversal $U(-\mathbf{p}) = U(\mathbf{p})^{-1}$, exchanges them (`inversion_swaps`, `U_parity`). With $Q = T_{3L} + T_{3R} + (B-L)/2$, one generation gets the right charges and $\sin^2\theta_W = 3/8$ at unification (`charges_leptons`, `charges_quarks`, `weinberg_full`).

A.2 The nine-sphere and Pati–Salam

Where do $B - L$ and colour come from? The round sphere $S^9 \subset \mathbb{R}^{10}$ has isometry group $SO(10)$. Split $\mathbb{R}^{10} = \mathbb{R}^4 \oplus \mathbb{R}^6$ and stretch the first factor by $\rho \neq 1$. Only matrices that do not mix the two factors survive (`ellipsoid_breaks`), leaving $SO(4) \times SO(6)$, locally $SU(2)_L \times SU(2)_R \times SU(4)_C$: the Pati–Salam group.

Theorem A.1. proved in Lean *The sixteen weights of the spinor representation of $SO(10)$, split as three plus two, carry the electric charge, $B - L$ and the two isospins of one Standard Model*

generation including a right-handed neutrino. The vector representation contains a Higgs bidoublet, and a Yukawa coupling $16 \cdot 16 \cdot 10$ *is allowed by the weights.*

In the repository: `spinor16_count`, `spinor16_charges`, `higgs_bidoublet`, `yukawa_weight`.

The direction of $B-L$ inside $SU(4)$ is fixed by colour symmetry and tracelessness, and its normalisation by the neutrality of the left-handed neutrino (`bl_direction`, `bl_normalization`). The $B-L$ anomaly cancels only if the right-handed neutrino is present (`bl_anomaly_cubic`, `bl_anomaly_needs_nuR`).

A.3 Chirality

The lowest Dirac modes of a round sphere are not zero modes: on S^9 the scalar curvature is 72, and the Lichnerowicz formula forces every harmonic spinor to vanish (`s9_scalar`, `no_zero_mode`). The index of the Dirac operator in odd dimensions is zero anyway (`index_zero`). So massless chiral fermions cannot come from the internal sphere alone. negative result

They can come from a domain wall. If a mass changes sign across a wall, there is exactly one normalisable zero mode, of one chirality (`jr_unique`, `kink_plus`, `norm_plus_lt`, `norm_minus`). This is the Jackiw–Rebbi mechanism, and Chapter 25 finds the same structure on the dumbbell: the mass W/r of the reduced fermion changes sign at the junction, and the zero mode sits there.

Where this stands

The group theory of $SO(10)$, Pati–Salam and the anomaly conditions are standard. What is specific here is the route: the quaternionic obstruction to $U(1)_Y$, its removal by left and right multiplication, and the ellipsoidal S^9 as the source of the breaking. The dynamics that would choose the ellipsoid, and the potential of the resulting Higgs field, are open; a curvature-only selection was tested and fails (Chapter 27). open

Exercises

Exercise A.1. Show that a pure quaternion commuting with i and j is zero. *Answer:* `center_pure_trivial`.

Exercise A.2. Show that a diagonal matrix commuting with the colour permutations has three equal entries. *Answer:* `bl_commute_color`.

Exercise A.3. Show that $\sum (B-L)^3 = 0$ over a generation with a right-handed neutrino and $\neq 0$ without it. *Answer:* `bl_anomaly_cubic`, `bl_anomaly_without_nuR`.

Exercise A.4. Show that $1/\cosh y$ is square integrable on the line and $\cosh y$ is not. *Answer:* `norm_plus_lt`, `norm_minus`.

Part III

Curved space and field equations

Part IV

The dumbbell

Chapter 10

The shape of the dumbbell

The question

Why can the cap and the neck not be joined smoothly? Because, as we will see, the neck is the envelope of the caps. Both solve the same first-order equation; the cap is its regular solution and the neck its singular one, and a curve that switches from a solution to the envelope of the solutions is C^1 but never C^2.

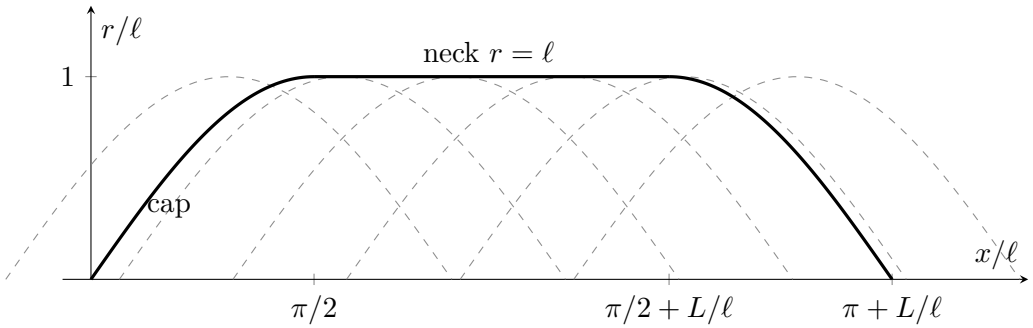


Figure 10.1: The areal radius along a time-symmetric slice of the dumbbell (solid). The dashed curves are other members of the family $r = \ell \sin((x - c)/\ell)$; the neck $r = \ell$ touches every one of them at its top. It is their envelope.

10.1 Kinematics: what a slice looks like

Freeze the dumbbell at a moment of time symmetry, when nothing is moving, and look at the space it occupies. Spherical symmetry lets us write its metric as

$$ds^2 = dx^2 + r(x)^2\, d\Omega^2, \tag{10.1}$$

where x is proper distance along the axis and $4\pi r^2$ is the area of the sphere at x; r is called the *areal radius*. The magnetic $SU(2)$ Yang–Mills field compatible with this symmetry is described by one function $W(x)$ (Chapter 12): its radial magnetic field is $(1 - W^2)/r^2$ and its tangential one is W'/r.

Two numbers attached to each of these spheres will carry most of the argument.

Definition 10.1. The *Misner–Sharp mass* of the sphere at x on a time-symmetric slice is $m = \frac{r}{2}(1 - r'^2)$. The *magnetic self-energy* of that sphere is $\ell^2(1 - W^2)^2/(2r)$.

For the cap $r = \ell \sin(x/\ell)$ one finds $m = \frac{\ell}{2}\sin^3(x/\ell)$, and for the neck $r = \ell$ one finds $m = \ell/2$.

The slice is *conformally flat* when its Weyl curvature vanishes. In this sector the relevant invariant can be computed in double null coordinates, and it comes out as a difference of the two numbers above.

Proposition 10.2 (Weyl curvature = mass − self-energy). proved in Lean *In spherically symmetric Einstein–Yang–Mills theory the conformally invariant combination of the metric functions equals*

$$3\Big(\frac{2m}{r} - \frac{\ell^2(1-W^2)^2}{r^2}\Big).$$

Hence the slice is conformally flat exactly when $2mr = \ell^2(1-W^2)^2$: on every sphere the mass equals the magnetic self-energy.

In the repository: `CFT.DS.ds_weyl_mass`.

10.2 The action and the constraint

The Einstein–Yang–Mills action of Chapter 13, reduced to spherical symmetry, gives one equation that every slice must satisfy, the Hamiltonian constraint. At a moment of time symmetry it reads

$$-4rr'' + 2(1 - r'^2) = 4\ell^2\Big(W'^2 + \frac{(1-W^2)^2}{2r^2}\Big), \qquad \ell^2 = \frac{4\pi G}{g^2}. \tag{10.2}$$

The left side is r^2 times the scalar curvature of the slice; the right side is r^2 times $16\pi G$ times the Yang–Mills energy density. You can check by hand that the cap $r = \ell\sin(x/\ell)$, $W = \cos(x/\ell)$ and the neck $r = \ell$, $W = 0$ both satisfy it: in both cases the two sides are $6\sin^2(x/\ell)$ and 2.

Multiplying (10.2) by $r'/4$ turns it into a statement about the mass:

$$m' = r'\,\ell^2\Big(W'^2 + \frac{(1-W^2)^2}{2r^2}\Big). \tag{10.3}$$

In the repository: `CFT.DS.ds2_mass_prime`. The mass grows wherever the sphere grows, at a rate fixed by the energy density.

10.3 A Bogomolny inequality

Now impose conformal flatness, $2mr = \ell^2(1-W^2)^2$, and differentiate it along the slice. Using (10.3) and dividing by $2\ell^2$ gives

$$rr'W'^2 + 2WW'(1-W^2) + \frac{(1-W^2)^2 r'}{r} = 0. \tag{10.4}$$

At first sight this is not illuminating. The trick, familiar from monopole physics, is to multiply by r' and look for squares.

Lemma 10.3 (completing the square). proved in Lean *For all real $r \neq 0$, r', W, W',*

$$r'\Big(rr'W'^2 + 2WW'(1-W^2) + \frac{(1-W^2)^2 r'}{r}\Big) = \frac{\big(rr'W' + W(1-W^2)\big)^2 + (1-W^2)^2\,(r'^2 - W^2)}{r}.$$

Lean: `ds2_bogomolny`

```
theorem ds2_bogomolny (r r' W W' : ℝ) (hr : r ≠ 0) :
    r' * (r * r' * W' ^ 2 + 2 * W * W' * (1 - W ^ 2)
          + (1 - W ^ 2) ^ 2 * r' / r)
      = ((r * r' * W' + W * (1 - W ^ 2)) ^ 2
          + (1 - W ^ 2) ^ 2 * (r' ^ 2 - W ^ 2)) / r := by
  field_simp
  ring
```

Theorem 10.4 (Bogomolny inequality for conformally flat slices)**.** proved in Lean *On a time-symmetric, conformally flat slice of spherically symmetric Einstein–Yang–Mills theory, at every point where $W^2 \neq 1$,*

$$r'^2 \leq W^2.$$

Equality holds if and only if $W = \pm r'$, and then the slice obeys the first-order equation $rr'W' = -W(1 - W^2)$.

Proof. By (10.4) the left side of the lemma vanishes, so the numerator on the right vanishes. It is a square plus $(1-W^2)^2(r'^2 - W^2)$ with $(1-W^2)^2 > 0$. If $r'^2 > W^2$ the numerator would be positive. At equality the second term is zero, so the square is zero. □

In the repository: `CFT.DS.ds2_num`, `ds2_W_ge`, `ds2_saturated`.

Remark. Earlier chapters imposed $W = r'$ by hand, motivated by the observation that the Yang–Mills potential can be identified with the spin connection of the slice (Chapter 15). Theorem 10.4 shows that this condition is not an extra assumption inside the conformally flat class: it is the case of equality of an inequality that every such slice satisfies. This is the same logic as the Bogomolny bound for monopoles, where the first-order equations describe the configurations that saturate an energy inequality.

10.4 The circle and its envelope

In the saturating case $W = r'$ we have $m = \frac{r}{2}(1 - W^2)$, and conformal flatness $2mr = \ell^2(1 - W^2)^2$ becomes $r^2(1 - W^2) = \ell^2(1 - W^2)^2$. Dividing by $1 - W^2$:

Theorem 10.5 (the shape equation)**.** proved in Lean *On a time-symmetric, conformally flat slice with $W = r'$ and $W^2 \neq 1$,*

$$r^2 + \ell^2 r'^2 = \ell^2. \tag{10.5}$$

In the repository: `CFT.DS.ds_circle_of_flat`.

Equation (10.5) says that the point $(r, \ell W) = (r, \ell r')$ moves on a circle of radius ℓ (Figure 10.2). The cap goes round half of it, from the north pole $(0, \ell)$ through the equator $(\ell, 0)$ to the south pole $(0, -\ell)$. The neck does not move at all: it sits at the single point $(\ell, 0)$.

Lean: `ds_cap_on_circle`

```
theorem ds_cap_on_circle (l x : ℝ) :
    (l * sin (x / l)) ^ 2 + l ^ 2 * cos (x / l) ^ 2 = l ^ 2 := by
  linear_combination l ^ 2 * sin_sq_add_cos_sq (x / l)
```

Equation (10.5) is of first order, and first-order equations can have solutions that are not members of the general family. Differentiating it gives $2r'(r + \ell^2 r'') = 0$, so at every point one of two things happens.

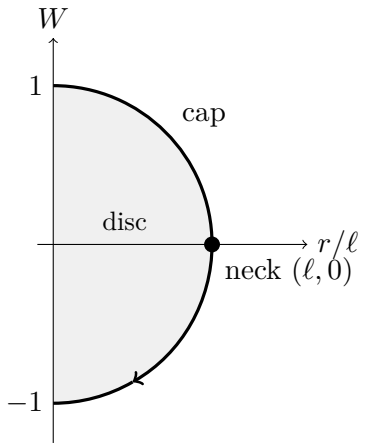


Figure 10.2: The (r, W) plane. Conformally flat slices with $W = r'$ live on the circle $r^2 + \ell^2 W^2 = \ell^2$. The cap runs along it; the neck is the point $(\ell, 0)$. The shaded disc is the region where conformally flat solutions have real null directions (Proposition 10.8).

Theorem 10.6 (two branches)**.** proved in Lean *Let r be twice differentiable and satisfy* (10.5)*. Then at every point*

$$r' = 0 \qquad \text{or} \qquad r'' = -\frac{r}{\ell^2}.$$

In the repository: `CFT.DS.ds_branch`.

The second branch is the harmonic equation; its solutions are the shifted caps $r = \ell \sin((x - c)/\ell)$, the regular solutions. The first branch, with $r^2 = \ell^2$, is the constant $r = \ell$: the neck. It is the *singular solution* of (10.5), the envelope of the family of caps (Figure 10.1). At $r = \ell$ the equation $r' = \pm\sqrt{\ell^2 - r^2}/\ell$ loses uniqueness, and this is the only place where a solution can leave the family of caps.

Corollary 10.7. *A C^1 curve that follows a cap and then the neck must switch at a point where the cap reaches $r = \ell$. There the first derivatives agree ($r' = 0$ on both sides) but the second derivatives are $-1/\ell$ and 0. The dumbbell is C^1 and never C^2.*

This is the geometric content of the "jump" that appeared in earlier chapters. It is not a defect put in by hand; it is the standard relation between a family of curves and its envelope: an envelope touches each curve, so slopes agree, but it does not bend with it.

Proposition 10.8 (the disc)**.** proved in Lean *For conformally flat solutions the slope dW/dr along a null direction solves a quadratic equation with coefficients $(1 - W^2)^2/r$, $2W(1 - W^2)$ and $(r^2 - \ell^2(1 - W^2)^2)/r$. Its discriminant is*

$$\frac{4(1 - W^2)^3}{r^2}\big(\ell^2(1 - W^2) - r^2\big),$$

so for $W^2 < 1$ real slopes exist exactly in the disc $r^2 + \ell^2 W^2 \le \ell^2$. The two roots coincide on the circle. The whole cap spacetime, $r = \ell \cos\eta \sin\chi$ and $W = \cos\chi$, lies in the disc, since $r^2 + \ell^2 W^2 = \ell^2 - \ell^2 \sin^2\eta \sin^2\chi$.

In the repository: `CFT.DS.ds_slope_disc`, `ds_slope_real`, `ds_cap_disc`.

10.5 Theorem A: the junction is forced

We can now put the pieces together. Once one decides to glue a cap to a neck, nothing about the gluing is left to choose.

Theorem 10.9 (uniqueness of the junction). proved in Lean *Consider the cap $r = \ell \sin(x/\ell)$ and the neck $r = \ell$ with $W = r'$.*

(i) The areal radius is continuous only at the equator $x = \pi\ell/2$ (`ju_location`).

(ii) There $r' = \cos(x/\ell) = 0$, so the first derivatives agree automatically (`ju_C1_auto`).

(iii) The second derivatives are $-1/\ell$ on the cap (`ju_cap_second`) and 0 on the neck, and at $r = \ell$, $r' = 0$ the Hamiltonian constraint (10.2) reduces to $r''(r'' + 1/\ell) = 0$, whose only roots are these two values (`rr_two_roots`).

(iv) The junction cannot be a timelike tube: along it the cap's equatorial radius $\sqrt{\ell^2 - \tau^2}$ differs from ℓ for $\tau \neq 0$ (`rj_tube_mismatch`).

(v) Along a null front the jump $[W_u]$ keeps its value at the corner, $1/(\sqrt{2}\,\ell)$ (`ju_jump_fixed`).

The combined statement is `CFT.JU.ju_theorem_A`.

Part (i) surprised the author. One would expect the place of the junction to be a free choice. It is not: the largest radius of the cap is exactly the radius of the neck, so they can touch only at the top of the cap, and there the slopes agree by themselves. "Glue continuously" already implies "glue in a C^1 way".

Where this stands

Known. The cap (the closed radiation universe driven by the homogeneous Yang–Mills field) and the neck (the magnetic Bertotti–Robinson spacetime) are classical solutions [1, 2, 3]. That the throat of an extremal black hole has a free length, and that it saturates an area–charge inequality, is known for Einstein–Maxwell theory [5, 6]. The relation between the Weyl curvature and the Misner–Sharp mass is known in general [4].

Claimed as new, in the weak sense of the guide for the specialist. The Bogomolny-type inequality $r'^2 \leq W^2$ for conformally flat slices (Theorem 10.4), the reading of the neck as the singular solution and envelope of the caps (Theorems 10.5 and 10.6), and the statement that the junction is forced (Theorem 10.9).

Assumptions to keep in mind. Spherical symmetry, a purely magnetic field, time symmetry and conformal flatness. Outside the class $W = r'$ the two roots of (iii) disappear, so these theorems do not exclude a smooth interpolation in general. Inside the conformally flat class a numerical search for a smooth static interpolation failed negative result (Chapter 19).

Open. Why the dumbbell is glued at all is not answered by these theorems; they say only how it must be glued if it is. Chapter 26 shows that a fixed total volume $2\pi^2\ell^3 + 4\pi\ell^2 L$ fixes the neck length L, and Chapter 23 shows that quantum mechanically L is conjugate to $-r\dot{r}$. open

Exercises

Each exercise is answered by the named Lean theorem. Read its statement in the repository and compare it with your own answer.

Exercise 10.1. Check by hand that the cap $r = \ell \sin(x/\ell)$, $W = \cos(x/\ell)$ lies on the circle (10.5), and that the neck $(r, W) = (\ell, 0)$ lies on it too. *Answer:* `ds_cap_on_circle`, `ds_neck_on_circle`.

Exercise 10.2. Multiply the Hamiltonian constraint (10.2) by $r'/4$ and show that it becomes (10.3). *Answer:* `ds2_mass_prime`.

Exercise 10.3. Expand both sides of the completing-the-square lemma and check that they agree. Where is $r \neq 0$ used? *Answer:* `ds2_bogomolny`.

Exercise 10.4. Theorem 10.5 assumes $W^2 \neq 1$. Find what goes wrong in the proof when $W^2 = 1$, and describe the slices with $W \equiv \pm 1$. *Answer:* the division in `ds_circle_of_flat`; for the slices, `fam_birkhoff_u` in Chapter 21.

Exercise 10.5. Show that $\ell \sin(x/\ell) = \ell$ with $0 < x < \pi\ell$ forces $x = \pi\ell/2$. *Answer:* `ju_location`.

Exercise 10.6. Compute the derivative of $\cos(x/\ell)$ at $x = \pi\ell/2$. *Answer:* `ju_cap_second`.

Exercise 10.7. Solve $2\ell^2 y^2 + 2\ell y = 0$ and explain why its two roots are the second derivatives of the neck and of the cap at the junction. *Answer:* `rr_two_roots`.

Exercise 10.8. Show that for $W^2 < 1$ the discriminant of Proposition 10.8 is non-negative if and only if $r^2 + \ell^2 W^2 \leq \ell^2$. *Answer:* `ds_slope_real`.

Part V

Heat and quanta

Part VI

Shape, swampland and open problems

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